Quantum Fourier Transform

Assessment of the QFT algorithm

QUESTION PAPER

Time allowed: 3 hours **Maximum marks:** 240
Questions: 80 (4 sections × 20) **Answer key:** separate Answer Book

Name of the candidate: _____ **Roll No.:** _____

General instructions

1. This question paper has **four sections** of **20 questions** each: **A** Mathematics for the QFT, **B** Quantum Mechanics, **C** QFT Basics, **D** Advanced QFT.
2. Every question is of the **single-correct** type: four options, exactly one correct. Question formats used: single-correct MCQ, matching lists (List-I/List-II), Assertion–Reason, Statement I/II and paragraph-based questions.
3. Mark your answers on the **Response Sheet** at the end of this paper by darkening ONE bubble per question.
4. Calculators are **not** allowed. Rough work may be done on separate sheets.
5. Unless a question says otherwise, use the **conventions** printed below; they apply to every section.
6. The detailed solutions are in the separate *Answer Book*; do not consult it before finishing the test.

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks: +3 if ONLY the correct option is chosen;
Zero marks: 0 if none of the options is chosen (question unanswered);
Negative marks: −1 in all other cases (wrong option, or more than one option chosen).

Conventions and notation used throughout the paper

- n qubits, $N = 2^n$, $\omega = e^{2\pi i/N}$. A basis state is written $|x\rangle = |x_1 x_2 \dots x_n\rangle$ with $x = \sum_j x_j 2^{n-j}$: **qubit 1 is the most significant bit** and is written first.
- $\mathrm{QFT}\,|x\rangle = \dfrac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega^{xy}\,|y\rangle$, $\quad \mathrm{QFT}^{-1}\,|y\rangle = \dfrac{1}{\sqrt{N}} \sum_{x=0}^{N-1} \omega^{-xy}\,|x\rangle$. Binary fraction: $0.x_l x_{l+1} \dots x_m = \frac{x_l}{2} + \frac{x_{l+1}}{4} + \dots + \frac{x_m}{2^{m-l+1}}$.
- $H = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$; X, Y, Z are the Pauli matrices; $R_k = \mathrm{diag}\left(1, e^{2\pi i/2^k}\right)$, so $R_1 = Z$, $S = R_2$, $T = R_3$; $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.
- **Textbook QFT circuit:** for $j = 1, \dots, n$: apply H to qubit j, then for $k = 2, \dots, n-j+1$ a controlled-R_k with control qubit $j+k-1$ and target qubit j; finally SWAP qubits j and $n+1-j$ for $j \le n/2$.
- **Phase estimation (QPE):** $U\,|u\rangle = e^{2\pi i\varphi}\,|u\rangle$, $0 \le \varphi < 1$; t counting qubits in $|+\rangle$ control $U^{2^{t-1}}, \dots, U^2, U$; the inverse QFT is applied and the outcome y gives the estimate $\varphi \approx y/2^t$.
- All measurements are in the computational basis unless stated otherwise; log means $\log_2$.

SECTION A · Mathematics for the QFT Q.1–20

Syllabus: complex numbers, roots of unity, geometric series, the discrete Fourier transform, binary fractions, unitary matrices, continued fractions and modular arithmetic.
Formats: Single-correct MCQ (Q.1–14); Matching lists (Q.15–16); Assertion–Reason (Q.17); Statement I / II (Q.18); Paragraph based (Q.19–20). Each question: +3 / 0 / −1.

Q.1 Let $\omega = e^{2\pi i/8}$. The value of $S = \sum_{k=0}^{7} k\,\omega^k$ is

(A) $\dfrac{8}{\omega - 1}$ **(B)** $\dfrac{8}{1-\omega}$ **(C)** 0 **(D)** $\dfrac{8\omega}{\omega - 1}$

Q.2 The number of ordered pairs (x, y) with $x, y \in \{0, 1, \ldots, 15\}$ for which $\sum_{z=0}^{15} e^{2\pi i\,(x+y)z/16} \neq 0$ is

(A) 1 **(B)** 16 **(C)** 15 **(D)** 31

Q.3 Let $x = 45$, written with $n = 6$ bits as $x_1x_2\ldots x_6$. The value of $e^{2\pi i\,x/2^4}$ is

(A) $e^{i\,13\pi/8}$ **(B)** $e^{i\,5\pi/8}$ **(C)** $e^{i\,3\pi/8}$ **(D)** $e^{-i\,13\pi/8}$

Q.4 The matrix $U = \dfrac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & c & -1 & d \end{pmatrix}$ is unitary. Then (c, d) equals

(A) $(i,\ -i)$ **(B)** $(1,\ -1)$ **(C)** $(-1,\ 1)$ **(D)** $(-i,\ i)$

Q.5 Let F be the 4×4 unitary Fourier matrix with entries $F_{yx} = \frac{1}{2}\,i^{xy}$ $(x, y = 0, \ldots, 3)$. The trace of F is

(A) $1 + i$ **(B)** 2 **(C)** 0 **(D)** $1 - i$

Q.6 Let $f(\theta) = \left|\,1 + e^{i\theta} + e^{2i\theta} + e^{3i\theta}\,\right|^2$. The maximum value of f and the smallest positive θ with $f(\theta) = 0$ are, respectively,

(A) 4 and $\pi/2$ **(B)** 4 and π **(C)** 16 and $\pi/4$ **(D)** 16 and $\pi/2$

Q.7 The best rational approximation p/q to $\dfrac{11}{64}$ with $1 \le q \le 10$ (as produced by the continued-fraction algorithm) is

(A) $2/9$ **(B)** $1/6$ **(C)** $1/7$ **(D)** $1/5$

Q.8 The order r of $a = 3$ modulo $N = 35$ (the least $r > 0$ with $a^r \equiv 1 \pmod{35}$), and the numbers $\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$, are

(A) $r = 6$; 5 and 7 **(B)** $r = 12$; 1 and 35
(C) $r = 12$; 7 and 5 **(D)** $r = 4$; 5 and 7

Q.9 A general pure state of 10 qubits is described by D complex amplitudes. After removing normalisation and the unobservable global phase, the number of independent *real* parameters is P. Then (D, P) is

(A) $(1024,\ 2048)$ **(B)** $(1024,\ 2046)$ **(C)** $(1024,\ 2047)$ **(D)** $(20,\ 40)$

Q.10 For $n = 4$, $H^{\otimes 4}|x\rangle = \frac{1}{4}\sum_y (-1)^{x\cdot y}|y\rangle$ where $x \cdot y = \sum_j x_j y_j$. For $x = 1011_2$, the number of $y \in \{0, \ldots, 15\}$ with a *negative* amplitude is

(A) 4 **(B)** 8 **(C)** 12 **(D)** 3

Q.11 A classical FFT on a vector of length 2^n needs about $n\,2^n$ operations, while the QFT circuit needs $\frac{n(n+1)}{2} + \lfloor n/2 \rfloor$ gates. For $n = 20$ the ratio (FFT operations)/(QFT gates) is closest to

(A) 10^9 **(B)** 10^3 **(C)** 10^5 **(D)** 10^7

Q.12 With $\omega = e^{2\pi i/8}$, the value of $\displaystyle\prod_{k=1}^{7} |1 - \omega^k|$ is

(A) 8 **(B)** 0 **(C)** 1 **(D)** 7

Q.13 The normalised vector $\alpha = \frac{1}{3}(1, 2, 2, 0)$ is transformed by the unitary 4×4 Fourier matrix $\beta_y = \frac{1}{2}\sum_x \alpha_x\, i^{xy}$. The value of $|\beta_0|^2$ is

(A) $\dfrac{5}{6}$ **(B)** $\dfrac{25}{12}$ **(C)** $\dfrac{1}{4}$ **(D)** $\dfrac{25}{36}$

Q.14 Let $\varphi = 0.\varphi_1\varphi_2\varphi_3$ (binary, $\varphi_j \in \{0, 1\}$). Then $e^{2\pi i\, 4\varphi}$ equals

(A) $e^{2\pi i\, 0.\varphi_2\varphi_3}$ **(B)** i^{φ_2} **(C)** $(-1)^{\varphi_1}$ **(D)** $(-1)^{\varphi_3}$

Q.15 Let $\omega = e^{2\pi i/8}$. Match List-I with List-II.

List-I		**List-II**	
(P)	ω^4	**(1)**	$\sqrt{2}$
(Q)	$\sum_{k=0}^{7} \omega^{2k}$	**(2)**	-1
(R)	$\omega \cdot \omega^7$	**(3)**	0
(S)	$\omega + \omega^7$	**(4)**	1

(A) P→1, Q→3, R→4, S→2 **(B)** P→2, Q→4, R→3, S→1
(C) P→2, Q→3, R→4, S→1 **(D)** P→3, Q→2, R→1, S→4

Q.16 F is the 4×4 unitary Fourier matrix ($F_{yx} = i^{xy}/2$), H the Hadamard matrix and $CT = \mathrm{diag}(1, 1, 1, e^{i\pi/4})$. Match each matrix in List-I with the *only* property in List-II that it has.

List-I		**List-II**	
(P)	F	**(1)**	a permutation matrix that is its own inverse
(Q)	F^2	**(2)**	real, symmetric, its own inverse, but not a permutation matrix
(R)	$H \otimes H$	**(3)**	diagonal with determinant $e^{i\pi/4}$
(S)	CT	**(4)**	$M^4 = I$ but $M^2 \neq I$

(A) P→4, Q→2, R→1, S→3 **(B)** P→4, Q→1, R→2, S→3
(C) P→3, Q→1, R→2, S→4 **(D)** P→1, Q→4, R→2, S→3

Q.17 **Assertion (A):** For every integer k that is not a multiple of N, $\sum_{j=0}^{N-1} e^{2\pi i\, jk/N} = 0$.

Reason (R): For every integer k that is not a multiple of N, the N numbers $e^{2\pi i\, jk/N}$ ($j = 0, \ldots, N-1$) are the N distinct vertices of a regular N-gon centred at the origin.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.18 **Statement I:** Every unitary matrix U satisfies $|\det U| = 1$.
Statement II: The unitary Fourier matrix F_N ($N = 2^n$, entries $\omega^{xy}/\sqrt{N}$) has $\det F_N = \pm 1$ for every n.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Paragraph for Questions 19 and 20

Let $N = 32$ and define the sequence $a_x = 1$ if $x \equiv 2 \pmod 5$ and $a_x = 0$ otherwise, for $x = 0, 1, \ldots, 31$. Its (un-normalised) Fourier coefficients are $A_y = \sum_{x=0}^{31} a_x\, e^{2\pi i\, xy/32}$, $y = 0, 1, \ldots, 31$. (The sequence is "periodic with period 5", but 5 does not divide 32.)

Q.19 The value of $\sum_{y=0}^{31} |A_y|^2$ is

(A) 192 **(B)** 6 **(C)** 36 **(D)** 1152

Q.20 Among $y = 1, 2, \ldots, 15$, the value of y for which $|A_y|$ is largest is

(A) 6
(B) 16 is the only peak; all of $1, \ldots, 15$ give $|A_y| = 0$
(C) 7
(D) 13

SECTION B · Quantum Mechanics Q.21–40

Syllabus: state vectors, Born rule, global and relative phases, gates and unitarity, tensor products, entanglement, partial measurement and phase kickback.
Formats: Single-correct MCQ (Q.21–34); Matching lists (Q.35–36); Assertion–Reason (Q.37); Statement I / II (Q.38); Paragraph based (Q.39–40). Each question: +3 / 0 / −1.

Q.21 A two-qubit state is $|\psi\rangle = c\,(\,|00\rangle + 2i\,|01\rangle - |10\rangle + \sqrt{2}\,|11\rangle\,)$ with $c > 0$. If only the *first* qubit is measured in the computational basis, the probability of obtaining 1 is

(A) $\frac{1}{2}$ **(B)** $\frac{3}{8}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{5}{8}$

Q.22 Which pair of single-qubit state vectors describes the *same* physical state (i.e. gives identical measurement statistics in every basis)?

(A) $|0\rangle + i\,|1\rangle$ and $i\,|0\rangle - |1\rangle$
(B) $|0\rangle + i\,|1\rangle$ and $i\,|0\rangle + |1\rangle$
(C) $|0\rangle + i\,|1\rangle$ and $|0\rangle - i\,|1\rangle$
(D) $|0\rangle + |1\rangle$ and $|0\rangle - |1\rangle$

Q.23 The state $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{i\phi}\,|1\rangle\,)$ passes through a Hadamard gate and is then measured in the computational basis. The probability of outcome 0 is

(A) $\cos^2(\phi/2)$ **(B)** $\sin^2(\phi/2)$ **(C)** $\frac{1}{2}$ **(D)** $\cos^2\phi$

Q.24 For the qubit state $|\psi\rangle = \cos\frac{\theta}{2}\,|0\rangle + e^{i\phi}\sin\frac{\theta}{2}\,|1\rangle$ the quantity $\langle X\rangle + i\langle Y\rangle$, where $\langle A\rangle = \langle\psi|\,A\,|\psi\rangle$ and X, Y are Pauli matrices, equals

(A) $\cos\theta\; e^{i\phi}$ **(B)** $\sin\theta\; e^{i\phi}$ **(C)** $\sin\frac{\theta}{2}\,e^{i\phi}$ **(D)** $\sin\theta\; e^{-i\phi}$

Q.25 Which of the following two-qubit states is *entangled* (cannot be written as $|a\rangle \otimes |b\rangle$)?

(A) $\frac{1}{2}(\,|00\rangle - |01\rangle + |10\rangle - |11\rangle\,)$
(B) $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle + |11\rangle\,)$
(C) $\frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle - |11\rangle\,)$
(D) $\frac{1}{2}(\,|00\rangle + i\,|01\rangle - |10\rangle - i\,|11\rangle\,)$

Q.26 A CNOT gate (qubit 1 = control, qubit 2 = target) acts on $|+\rangle\,|-\rangle$, where $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. The output is

(A) $|+\rangle\,|-\rangle$
(B) $|-\rangle\,|+\rangle$
(C) $\frac{1}{\sqrt{2}}(\,|0\rangle\,|-\rangle + |1\rangle\,|+\rangle\,)$
(D) $|-\rangle\,|-\rangle$

Q.27 In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ (first qubit = control), the matrix of controlled-R_3, where $R_k = \mathrm{diag}\,(1, e^{2\pi i/2^k})$, is

(A) $\mathrm{diag}\,(1,1,1,e^{i\pi/4})$
(B) $\mathrm{diag}\,(1,e^{i\pi/4},1,e^{i\pi/4})$
(C) $\mathrm{diag}\,(1,1,e^{i\pi/4},e^{i\pi/4})$
(D) $\mathrm{diag}\,(1,1,1,e^{i\pi/8})$

Q.28 $U\,|u\rangle = e^{2\pi i\varphi}\,|u\rangle$. A control qubit in $|+\rangle$ and the register in $|u\rangle$ pass through a controlled-U^{2^j} gate. The output state is

(A) $\frac{1}{\sqrt{2}}\,e^{2\pi i\,2^j\varphi}(\,|0\rangle + |1\rangle\,)\,|u\rangle$
(B) $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\,2^j\varphi}\,|1\rangle\,)\,|u\rangle$
(C) $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\,j\varphi}\,|1\rangle\,)\,|u\rangle$
(D) $\frac{1}{\sqrt{2}}(\,|0\rangle + e^{2\pi i\,\varphi/2^j}\,|1\rangle\,)\,|u\rangle$

Q.29 The state $\frac{1}{\sqrt{3}}(\,|00\rangle + |01\rangle + |11\rangle\,)$ is prepared and only the *first* qubit is measured, giving 0. The probability of this outcome and the post-measurement state of the second qubit are

(A) $\frac{2}{3}$ and $|+\rangle$ **(B)** $\frac{1}{3}$ and $|0\rangle$ **(C)** $\frac{2}{3}$ and $|0\rangle$ **(D)** $\frac{1}{3}$ and $|+\rangle$

Q.30 Which of the following matrices can *not* be a quantum gate?

(A) $\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ i & i\end{pmatrix}$ **(B)** $\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$ **(C)** $\frac{1}{\sqrt{2}}\begin{pmatrix}1 & i\\ i & 1\end{pmatrix}$ **(D)** $\begin{pmatrix}0 & 1\\ i & 0\end{pmatrix}$

Q.31 Which two-qubit gate does *not* commute with controlled-$R_2 = \mathrm{diag}(1,1,1,i)$?

(A) $Z \otimes Z$ **(B)** $S \otimes I$, where $S = \mathrm{diag}(1, i)$
(C) SWAP **(D)** $H \otimes I$

Q.32 A state-vector simulator stores each amplitude as a single-precision complex number (two 4-byte floats). The memory needed for the full state of 32 qubits is

(A) 256 bytes **(B)** 32 GiB **(C)** 16 GiB **(D)** 64 GiB

Q.33 $H \otimes H$ is applied to the Bell state $\frac{1}{\sqrt{2}}(\,|00\rangle - |11\rangle\,)$, followed by a measurement of both qubits. The probability of the outcome 01 is

(A) 0 **(B)** $\frac{1}{4}$ **(C)** $\frac{1}{2}$ **(D)** 1

Q.34 A source emits a qubit that is $|0\rangle$ with probability $\frac{1}{2}$ and $|1\rangle$ with probability $\frac{1}{2}$ (a classical coin decides; nobody knows which). A Hadamard gate is applied and the qubit is measured. The probability of outcome 0 is

(A) 1 **(B)** $\frac{1}{2}$ **(C)** $\frac{1}{4}$ **(D)** 0

Q.35 Match each single-qubit gate in List-I with its action in List-II. ($R_k = \mathrm{diag}(1, e^{2\pi i/2^k})$.)

List-I		**List-II**	
(P)	X	**(1)**	$\lvert 1\rangle \mapsto \frac{1}{\sqrt{2}}(\,\lvert 0\rangle - \lvert 1\rangle\,)$
(Q)	H	**(2)**	$\lvert 1\rangle \mapsto e^{i\pi/4}\lvert 1\rangle$
(R)	R_2	**(3)**	$\lvert 0\rangle \leftrightarrow \lvert 1\rangle$
(S)	R_3	**(4)**	$\lvert 1\rangle \mapsto i\lvert 1\rangle$

(A) P→3, Q→1, R→2, S→4 **(B)** P→1, Q→3, R→4, S→2
(C) P→3, Q→1, R→4, S→2 **(D)** P→4, Q→1, R→3, S→2

Q.36 Match each two-qubit state in List-I with its statistics when *both* qubits are measured in the computational basis (List-II).

List-I		**List-II**	
(P)	$\frac{1}{\sqrt{2}}(\,\lvert 00\rangle + \lvert 11\rangle\,)$	**(1)**	all four outcomes, each with probability $\frac{1}{4}$
(Q)	$\frac{1}{\sqrt{2}}(\,\lvert 01\rangle - \lvert 10\rangle\,)$	**(2)**	the two bits are always equal; 00 and 11 each $\frac{1}{2}$
(R)	$\frac{1}{2}(\,\lvert 0\rangle + i\,\lvert 1\rangle\,)(\,\lvert 0\rangle - \lvert 1\rangle\,)$	**(3)**	first bit always 0; second bit 0 or 1 with $\frac{1}{2}$ each
(S)	$\lvert 0\rangle\,\lvert +\rangle$	**(4)**	the two bits always differ; 01 and 10 each $\frac{1}{2}$

(A) P→1, Q→4, R→2, S→3 **(B)** P→4, Q→2, R→1, S→3

(C) P→2, Q→4, R→3, S→1 **(D)** P→2, Q→4, R→1, S→3

Q.37 **Assertion (A):** The states $|+\rangle$ and $|-\rangle$ cannot be distinguished by a measurement in the computational basis $\{|0\rangle, |1\rangle\}$.

Reason (R): $|+\rangle$ and $|-\rangle$ differ only by a global phase.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.38 **Statement I:** The Bell state $\frac{1}{\sqrt{2}}(\,\lvert 00\rangle + \lvert 11\rangle\,)$ can be written as $|a\rangle \otimes |b\rangle$ for suitable single-qubit states $|a\rangle$, $|b\rangle$.

Statement II: A two-qubit state $c_{00}\,|00\rangle + c_{01}\,|01\rangle + c_{10}\,|10\rangle + c_{11}\,|11\rangle$ is a product state if and only if $c_{00}c_{11} = c_{01}c_{10}$.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Paragraph for Questions 39 and 40

Hadamard test. A control qubit starts in $|0\rangle$ and a target register starts in an eigenstate $|u\rangle$ of a unitary U with $U|u\rangle = e^{2\pi i\varphi}|u\rangle$, $0 \le \varphi < 1$. The circuit applies, in order: (i) H on the control; (ii) controlled-U (control qubit $\to$ target register); (iii) H on the control; (iv) a measurement of the control qubit in the computational basis. This is the one-counting-qubit version of quantum phase estimation.

Q.39 The probability that the control qubit is found in state $|1\rangle$ is

(A) $\cos^2(\pi\varphi)$ **(B)** $\sin^2(\pi\varphi/2)$ **(C)** $\sin^2(2\pi\varphi)$ **(D)** $\sin^2(\pi\varphi)$

Q.40 Take $\varphi = \frac{1}{3}$. The circuit is run repeatedly (fresh $|0\rangle$ control each time). The minimum number of runs needed so that the probability of observing outcome 1 *at least once* exceeds 0.99 is

(A) 2 **(B)** 3 **(C)** 4 **(D)** 7

SECTION C · QFT Basics Q.41–60

Syllabus: definition and matrix of the QFT, product formula, textbook circuit, gate counts, inverse QFT and simple inputs.
Formats: Single-correct MCQ (Q.41–54); Matching lists (Q.55–56); Assertion–Reason (Q.57); Statement I / II (Q.58); Paragraph based (Q.59–60). Each question: +3 / 0 / −1.

Q.41 For $n = 2$, the amplitude of $|10\rangle$ in QFT $|11\rangle$ is

(A) $-\frac{1}{2}$ **(B)** $\frac{1}{2}$ **(C)** $-\frac{i}{2}$ **(D)** $\frac{i}{2}$

Q.42 A 5-qubit QFT is applied to the basis state $|x\rangle$ with $x = 19$ and the output is measured. The probability of obtaining $y = 7$ is

(A) $\frac{7}{32}$ **(B)** $\frac{1}{\sqrt{32}}$ **(C)** 0 **(D)** $\frac{1}{32}$

Q.43 For $n = 3$, QFT $|110\rangle$ equals (qubit 1 written first)

(A) $\frac{1}{\sqrt{8}}(|0\rangle - |1\rangle)(|0\rangle - |1\rangle)(|0\rangle - i|1\rangle)$ **(B)** $\frac{1}{\sqrt{8}}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle)(|0\rangle + i|1\rangle)$
(C) $\frac{1}{\sqrt{8}}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle)(|0\rangle - i|1\rangle)$ **(D)** $\frac{1}{\sqrt{8}}(|0\rangle - i|1\rangle)(|0\rangle - |1\rangle)(|0\rangle + |1\rangle)$

Q.44 The textbook QFT circuit on $n = 12$ qubits contains how many controlled-R_k gates?

(A) 144 **(B)** 72 **(C)** 66 **(D)** 78

Q.45 On hardware a SWAP is implemented with 3 CNOTs. Counting every controlled-R_k as one two-qubit gate, the total number of two-qubit gates in the 12-qubit textbook QFT (including the final SWAPs) is

(A) 72 **(B)** 84 **(C)** 102 **(D)** 66

Q.46 The smallest rotation angle appearing in the textbook QFT circuit on 6 qubits is

(A) 11.25° **(B)** 6° **(C)** 2.8125° **(D)** 5.625°

Q.47 In the textbook QFT circuit acting on $|x_1 x_2 \dots x_n\rangle$, *before* the final SWAPs, qubit 1 is in the state $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\phi}|1\rangle)$ with ϕ equal to

(A) $0.x_1 x_2 \dots x_n$ **(B)** $0.x_n x_{n-1} \dots x_1$ **(C)** $0.x_n$ **(D)** $0.x_1$

Q.48 For any n-qubit basis state, $\mathrm{QFT}^2 |x\rangle$ equals ($N = 2^n$)

(A) $|x\rangle$ **(B)** $|2x \bmod N\rangle$ **(C)** $|(-x) \bmod N\rangle$ **(D)** $|N - 1 - x\rangle$

Q.49 The QFT is applied to the uniform superposition $\frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$. The output is

(A) $|N - 1\rangle$ **(B)** $|0\rangle$
(C) the uniform superposition itself **(D)** $|N/2\rangle$

Q.50 For n qubits, $\mathrm{QFT}\,|10\ldots0\rangle$ (i.e. $x = 2^{n-1}$) equals

(A) $|+\rangle^{\otimes(n-1)} \otimes |-\rangle$ **(B)** $|-\rangle^{\otimes n}$ **(C)** $|+\rangle^{\otimes n}$ **(D)** $|-\rangle \otimes |+\rangle^{\otimes(n-1)}$

Q.51 The inverse QFT on 3 qubits is built by reversing the textbook QFT circuit and conjugating every gate. Ignoring the SWAP, the *first* gate of the inverse circuit is

(A) H on qubit 3
(B) controlled-$R_3^\dagger$ between qubits 1 and 3
(C) controlled-$R_2^\dagger$ between qubits 1 and 2
(D) H on qubit 1

Q.52 For $n = 4$, the phase (in degrees, in $[0°, 360°)$) of the amplitude of $|y\rangle$, $y = 3$, in $\mathrm{QFT}\,|x\rangle$, $x = 5$, is

(A) $22.5°$ **(B)** $337.5°$ **(C)** $168.75°$ **(D)** $15°$

Q.53 NumPy's two transforms of a length-N vector a are

$$\texttt{fft}(a)_y = \sum_x a_x\, e^{-2\pi i\, xy/N}, \qquad \texttt{ifft}(a)_y = \frac{1}{N}\sum_x a_x\, e^{+2\pi i\, xy/N}.$$

If $\mathrm{QFT}\sum_x \alpha_x\,|x\rangle = \sum_y \beta_y\,|y\rangle$ (convention box), then the vector β equals

(A) $\texttt{fft}(\alpha)$ **(B)** $\sqrt{N}\cdot\texttt{ifft}(\alpha)$ **(C)** $\texttt{ifft}(\alpha)/\sqrt{N}$ **(D)** $\texttt{fft}(\alpha)/\sqrt{N}$

Q.54 For an n-qubit computational basis state $|x\rangle$, the state $\mathrm{QFT}\,|x\rangle$ is

(A) a product state only when x is even
(B) always maximally entangled
(C) entangled unless $x = 0$
(D) always a product (unentangled) state

Q.55 Match the number of qubits n (List-I) with the number of controlled-R_k gates in the textbook QFT (List-II).

List-I		**List-II**	
(P)	$n = 4$	**(1)**	15
(Q)	$n = 6$	**(2)**	28
(R)	$n = 8$	**(3)**	45
(S)	$n = 10$	**(4)**	6

(A) P→1, Q→4, R→2, S→3
(B) P→4, Q→2, R→1, S→3
(C) P→4, Q→1, R→2, S→3
(D) P→4, Q→1, R→3, S→2

Q.56 For $n = 2$, match each input (List-I) with QFT of it (List-II); qubit 1 is written first.

List-I		**List-II**	
(P)	$\|00\rangle$	**(1)**	$\frac{1}{2}(\,|0\rangle - |1\rangle\,)(\,|0\rangle + i\,|1\rangle\,)$
(Q)	$\|01\rangle$	**(2)**	$\frac{1}{2}(\,|0\rangle - |1\rangle\,)(\,|0\rangle - i\,|1\rangle\,)$
(R)	$\|10\rangle$	**(3)**	$\frac{1}{2}(\,|0\rangle + |1\rangle\,)(\,|0\rangle + |1\rangle\,)$
(S)	$\|11\rangle$	**(4)**	$\frac{1}{2}(\,|0\rangle + |1\rangle\,)(\,|0\rangle - |1\rangle\,)$

(A) P→3, Q→4, R→1, S→2
(B) P→4, Q→1, R→3, S→2
(C) P→3, Q→2, R→4, S→1
(D) P→3, Q→1, R→4, S→2

Q.57 **Assertion (A):** Measuring QFT $|x\rangle$ in the computational basis reveals no information about the basis state x.

Reason (R): All N amplitudes of QFT $|x\rangle$ have the same magnitude $1/\sqrt{N}$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.58 **Statement I:** The QFT on n qubits can be implemented with $O(n^2)$ one- and two-qubit gates.

Statement II: Consequently, for an arbitrary classical input vector of length 2^n, a quantum computer can output all 2^n of its Fourier coefficients using only $O(n^2)$ operations in total.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Paragraph for Questions 59 and 60

The textbook 3-qubit QFT circuit is applied to the input $|x_1x_2x_3\rangle = |011\rangle$ ($x = 3$). The gates, in order, are: H on qubit 1; controlled-R_2 (control qubit 2, target qubit 1); controlled-R_3 (control qubit 3, target qubit 1); H on qubit 2; controlled-R_2 (control qubit 3, target qubit 2); H on qubit 3; SWAP of qubits 1 and 3. Here $R_k = \mathrm{diag}(1, e^{2\pi i/2^k})$, and "the phase of qubit j" means the angle θ in its state $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\theta}|1\rangle)$, taken in $[0°, 360°)$.

Q.59 Immediately after the controlled-R_3 gate (the third gate), the phase of qubit 1 is

(A) 270° **(B)** 135° **(C)** 90° **(D)** 45°

Q.60 After the complete circuit (including the SWAP), the amplitude of the basis state $|110\rangle$ is

(A) $\frac{i}{8}$ **(B)** $\frac{e^{i\pi/4}}{\sqrt{8}}$ **(C)** $\frac{-i}{\sqrt{8}}$ **(D)** $\frac{i}{\sqrt{8}}$

SECTION D · Advanced QFT Q.61–80

Syllabus: periodic states, shift theorem, phase estimation accuracy, Shor's algorithm, approximate and semiclassical QFT, spectral properties and applications.
Formats: Single-correct MCQ (Q.61–74); Matching lists (Q.75–76); Assertion–Reason (Q.77); Statement I / II (Q.78); Paragraph based (Q.79–80). Each question: +3 / 0 / −1.

Q.61 On $n = 6$ qubits ($N = 64$) the state $\frac{1}{\sqrt{8}} \sum_{j=0}^{7} |3 + 8j\rangle$ is prepared, the QFT is applied and all qubits are measured. The possible outcomes y and their probabilities are

(A) only $y = 8$, with probability 1
(B) all 64 values of y, each with probability $\frac{1}{64}$
(C) $y \in \{3, 11, 19, \ldots, 59\}$, each with probability $\frac{1}{8}$
(D) $y \in \{0, 8, 16, \ldots, 56\}$, each with probability $\frac{1}{8}$

Q.62 For the state $\frac{1}{\sqrt{8}} \sum_{j=0}^{7} |3 + 8j\rangle$ on 6 qubits, the phase (in $[0°, 360°)$) of the amplitude of $|16\rangle$ after the QFT is

(A) $90°$ **(B)** $270°$ **(C)** $135°$ **(D)** $0°$

Q.63 Let $\mathrm{QFT} \sum_x \alpha_x |x\rangle = \sum_y \beta_y |y\rangle$ on n qubits ($N = 2^n$, $\omega = e^{2\pi i/N}$). For a fixed integer s, $\mathrm{QFT} \sum_x \alpha_x |(x + s) \bmod N\rangle$ equals

(A) $\sum_y \omega^{-sy} \beta_y |y\rangle$ **(B)** $\sum_y \beta_{(y+s) \bmod N} |y\rangle$
(C) $\sum_y \beta_y |(y + s) \bmod N\rangle$ **(D)** $\sum_y \omega^{sy} \beta_y |y\rangle$

Q.64 To obtain φ accurate to n bits with probability at least $1 - \varepsilon$, phase estimation uses $t = n + \lceil \log_2(2 + \frac{1}{2\varepsilon}) \rceil$ counting qubits. For $n = 5$ and $\varepsilon = 0.1$, t equals

(A) 7 **(B)** 8 **(C)** 10 **(D)** 9

Q.65 Phase estimation with $t = 3$ counting qubits (inverse QFT readout) is run on an eigenstate with $\varphi = \frac{5}{8}$. The probability that the counting register reads 101 is

(A) $\frac{5}{8}$ **(B)** 1 **(C)** $\frac{1}{8}$ **(D)** $\frac{4}{\pi^2}$

Q.66 If φ is not exactly a t-bit binary fraction, phase estimation returns the *single* closest t-bit approximation with probability at least

(A) $\frac{1}{2}$ **(B)** $\frac{1}{\pi} \approx 0.318$ **(C)** $\frac{4}{\pi^2} \approx 0.405$ **(D)** $\frac{8}{\pi^2} \approx 0.81$

Q.67 Phase estimation with $t = 2$ counting qubits is run with $\varphi = \frac{1}{3}$. The probability of the outcome $y = 1$ (i.e. the estimate $\varphi \approx \frac{1}{4}$) is

(A) $\frac{6 - 3\sqrt{3}}{16}$ **(B)** $\frac{1}{3}$ **(C)** $\frac{4}{\pi^2}$ **(D)** $\frac{6 + 3\sqrt{3}}{16}$

Q.68 For Shor's algorithm with a work register of $n = \lceil \log_2 N \rceil$ qubits and a counting register of $2n$ qubits, the total number of qubits needed to factor $N = 21$ is

(A) 15 **(B)** 12 **(C)** 21 **(D)** 10

Q.69 Order finding for $a = 7$, $N = 15$ uses a counting register of $t = 8$ qubits ($M = 256$). The set of possible measurement outcomes y is

(A) $\{0, 128\}$ **(B)** $\{0, 64, 128, 192\}$ **(C)** $\{0, 4, 8, 12\}$ **(D)** $\{0, 85, 171\}$

Q.70 In the setting of the previous question (order finding for $a = 7$, $N = 15$, $M = 256$), the continued-fraction step turns $y/256$ into k/r'. Which outcome leads to the *wrong* candidate $r' = 2$?

(A) $y = 64$ **(B)** $y = 128$ **(C)** $y = 0$ **(D)** $y = 192$

Q.71 An approximate QFT on $n = 20$ qubits keeps only the controlled-R_k gates with $k \le m = 6$ and drops the rest. The number of controlled rotations dropped is

(A) 114 **(B)** 85 **(C)** 95 **(D)** 105

Q.72 On $n = 6$ qubits ($N = 64$) the uniform superposition of all $x \equiv 0 \pmod 6$, $0 \le x \le 63$, is prepared and the QFT applied. The probability of the outcome $y = 32$ is

(A) $\frac{11}{64}$ **(B)** $\frac{1}{64}$ **(C)** $\frac{1}{6}$ **(D)** $\frac{1}{11}$

Q.73 Every eigenvalue λ of the n-qubit QFT matrix ($n \ge 1$) satisfies

(A) $\lambda^N = 1$, and all N of the N-th roots of unity occur

(B) $|\lambda| < 1$ for at least one eigenvalue

(C) $\lambda \in \{1, -1\}$

(D) $\lambda \in \{1, -1, i, -i\}$

Q.74 In phase estimation the inverse QFT is immediately followed by measurement of every counting qubit. Using the semiclassical (Griffiths–Niu) QFT, the number of *two-qubit* gates needed for this inverse QFT on n qubits is

(A) $n - 1$ **(B)** $\frac{n(n-1)}{2}$ **(C)** 0 **(D)** n

Q.75 Match each algorithm in List-I with the role played by the QFT, or the related step, in List-II.

List-I		**List-II**	
(P)	Quantum phase estimation	**(1)**	adds a number by applying controlled phase rotations to a register held in the Fourier basis
(Q)	Shor's factoring algorithm	**(2)**	turns the phases $e^{2\pi i\, 2^j \varphi}$ in the counting register into the binary digits of φ
(R)	Draper's quantum adder	**(3)**	after a QPE step, applies an ancilla rotation proportional to $1/\lambda$ for each eigenvalue λ
(S)	HHL linear-system solver	**(4)**	extracts an order r from peaks near kM/r using continued fractions

(A) P→2, Q→4, R→1, S→3 **(B)** P→4, Q→2, R→1, S→3

(C) P→2, Q→1, R→4, S→3 **(D)** P→2, Q→4, R→3, S→1

Q.76 On $n = 4$ qubits ($N = 16$) the QFT is applied to each input in List-I and the result measured. Match with List-II.

List-I		**List-II**	
(P)	$\lvert 0\rangle$	**(1)**	only $y = 0$ and $y = 8$, each with probability $\frac{1}{2}$
(Q)	$\frac{1}{2}(\lvert 0\rangle + \lvert 4\rangle + \lvert 8\rangle + \lvert 12\rangle)$	**(2)**	all 16 outcomes, each with probability $\frac{1}{16}$
(R)	$\frac{1}{\sqrt{8}} \sum_{x \text{ even}} \lvert x\rangle$	**(3)**	only $y = 0, 4, 8, 12$, each with probability $\frac{1}{4}$
(S)	$\frac{1}{4} \sum_{x=0}^{15} \lvert x\rangle$	**(4)**	$y = 0$ with certainty

(A) P→4, Q→3, R→1, S→2
(B) P→2, Q→3, R→4, S→1
(C) P→2, Q→3, R→1, S→4
(D) P→2, Q→1, R→3, S→4

Q.77 **Assertion (A):** For large n, dropping every controlled-R_k with $k > m$, where m is only slightly larger than $\log_2 n$, changes the output of the QFT circuit only slightly.

Reason (R): Each dropped gate differs from the identity only by a phase $2\pi/2^k$, so its error is at most about $2\pi/2^k$; at most n gates are dropped for each k, giving a total error of order $n \cdot 2\pi/2^m$, which is small once $2^m \gg n$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.78 **Statement I:** For $\varphi = \frac{1}{3}$, phase estimation with any finite number t of counting qubits never returns an outcome y with $y/2^t = \varphi$ exactly, with certainty or otherwise.

Statement II: The number $\frac{1}{3}$ has no terminating binary expansion ($\frac{1}{3} = 0.010101\ldots_2$).

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Paragraph for Questions 79 and 80

Shor's algorithm for $N = 21$ with $a = 2$. A counting register of $t = 10$ qubits ($M = 2^{10} = 1024$) and a 5-qubit work register initialised to $|1\rangle$ are used. Steps: (i) $H^{\otimes 10}$ on the counting register; (ii) $|x\rangle\,|1\rangle \mapsto |x\rangle\,|2^x \bmod 21\rangle$; (iii) inverse QFT on the counting register; (iv) measure the counting register, obtaining y. Classical post-processing replaces $y/1024$ by the closest fraction k/r with $r < 21$ (continued fractions) and keeps r if $2^r \equiv 1 \pmod{21}$. The true order of 2 modulo 21 is 6, since $2^6 = 64 = 3 \cdot 21 + 1$.

Q.79 Which of the following measured values y leads the post-processing to the correct order $r = 6$?

(A) 683 **(B)** 512 **(C)** 171 **(D)** 341

Q.80 The probability that the measured value is exactly $y = 512$ is closest to

(A) 0.333 **(B)** 0.500 **(C)** 0.167 **(D)** 0.001

RESPONSE SHEET

Darken exactly ONE bubble per question. Leave all four bubbles empty to skip a question (0 marks).

Name: _____ **Roll No.:** _____

Section A	Section B	Section C	Section D
1 (A)(B)(C)(D)	21 (A)(B)(C)(D)	41 (A)(B)(C)(D)	61 (A)(B)(C)(D)
2 (A)(B)(C)(D)	22 (A)(B)(C)(D)	42 (A)(B)(C)(D)	62 (A)(B)(C)(D)
3 (A)(B)(C)(D)	23 (A)(B)(C)(D)	43 (A)(B)(C)(D)	63 (A)(B)(C)(D)
4 (A)(B)(C)(D)	24 (A)(B)(C)(D)	44 (A)(B)(C)(D)	64 (A)(B)(C)(D)
5 (A)(B)(C)(D)	25 (A)(B)(C)(D)	45 (A)(B)(C)(D)	65 (A)(B)(C)(D)
6 (A)(B)(C)(D)	26 (A)(B)(C)(D)	46 (A)(B)(C)(D)	66 (A)(B)(C)(D)
7 (A)(B)(C)(D)	27 (A)(B)(C)(D)	47 (A)(B)(C)(D)	67 (A)(B)(C)(D)
8 (A)(B)(C)(D)	28 (A)(B)(C)(D)	48 (A)(B)(C)(D)	68 (A)(B)(C)(D)
9 (A)(B)(C)(D)	29 (A)(B)(C)(D)	49 (A)(B)(C)(D)	69 (A)(B)(C)(D)
10 (A)(B)(C)(D)	30 (A)(B)(C)(D)	50 (A)(B)(C)(D)	70 (A)(B)(C)(D)
11 (A)(B)(C)(D)	31 (A)(B)(C)(D)	51 (A)(B)(C)(D)	71 (A)(B)(C)(D)
12 (A)(B)(C)(D)	32 (A)(B)(C)(D)	52 (A)(B)(C)(D)	72 (A)(B)(C)(D)
13 (A)(B)(C)(D)	33 (A)(B)(C)(D)	53 (A)(B)(C)(D)	73 (A)(B)(C)(D)
14 (A)(B)(C)(D)	34 (A)(B)(C)(D)	54 (A)(B)(C)(D)	74 (A)(B)(C)(D)
15 (A)(B)(C)(D)	35 (A)(B)(C)(D)	55 (A)(B)(C)(D)	75 (A)(B)(C)(D)
16 (A)(B)(C)(D)	36 (A)(B)(C)(D)	56 (A)(B)(C)(D)	76 (A)(B)(C)(D)
17 (A)(B)(C)(D)	37 (A)(B)(C)(D)	57 (A)(B)(C)(D)	77 (A)(B)(C)(D)
18 (A)(B)(C)(D)	38 (A)(B)(C)(D)	58 (A)(B)(C)(D)	78 (A)(B)(C)(D)
19 (A)(B)(C)(D)	39 (A)(B)(C)(D)	59 (A)(B)(C)(D)	79 (A)(B)(C)(D)
20 (A)(B)(C)(D)	40 (A)(B)(C)(D)	60 (A)(B)(C)(D)	80 (A)(B)(C)(D)

Score computation (filled in by the evaluator, using the Answer Book)

Sec.	**Topic**	**Attempted**	**Correct** C	**Wrong** W	**Marks** $3C - W$
A	Mathematics for the QFT				
B	Quantum Mechanics				
C	QFT Basics				
D	Advanced QFT				
				Total (out of 240)	